

CW2 Marking Criteria Response

Project: Smart Classroom Energy-Saving, Safety & Asset Monitoring System

GitHub: <https://github.com/3351666087/desktop-tutorial>

This document maps the delivered project directly to the IOT104TC Coursework 2 marking scheme.

1. Presentation - 20 Marks

The marking scheme awards 18-20 marks for an outstanding presentation of the idea, design, functions and operations.

Requirement	Evidence delivered
Clear idea	The project solves classroom energy waste, safety alarm response and facility monitoring.
Clear design	The slide deck and report explain physical, edge, gateway, service, intelligence and user layers.
Clear functions	Lighting, cooling, safety alarm, manual control, mobile voice command, analytics and presets are demonstrated.
Clear operations	The system includes live dashboard, MQTT logs, ACK feedback, status lights, fan RPM and physical actuator response.
Presentation materials	Final PPTX, PPT PDF, bilingual script, four-person presentation plan and feature guide are included.

Why this targets the top band: The presentation materials are not only descriptive. They are supported by a working physical system, a dashboard, a GitHub repository and a repeatable demo path.

2. Technical Knowledge And Operation - 50 Marks

The marking scheme awards 40-50 marks for a perfect technical presentation and operation beyond the design and goal of the group project.

Technical area	Project evidence
Hardware knowledge	Pin-level wiring for UNO, ESP32, sensors, relay, fan, status lights and UART level shifting.
Software knowledge	Arduino firmware, ESP32 gateway firmware, Node.js dashboard, MQTT broker, WebSocket live state, Python AI backend.
Architecture	Real layered architecture: physical layer, edge control, gateway, service layer, intelligence layer and user layer.
Security and privacy	local-first safety, ignored API keys, temporary voice files, validated command schema, LAN-only control prototype.
Operations	startup script, self-check script, compile checks, runtime AI check, dashboard health endpoints and onsite wiring checklist.
Demonstration depth	Mobile voice command changes the physical fan through ASR, Qwen, MQTT, ESP32 UART and Arduino control.

Technical area	Project evidence
Robustness	UNO remains responsible for safety and actuator control even if dashboard or cloud AI is unavailable.
Evidence	PASS 57 / WARN 0 / FAIL 0 , 11/11 Arduino/ESP32 sketches compiled, AI runtime verified on CUDA.

Why this targets the top band: The system goes beyond a basic Arduino lab task. It integrates embedded control, network gateway, MQTT service architecture, mobile web interaction, AI planning and physical actuation in one demonstrable system.

3. Practical Use-Cases - 30 Marks

The marking scheme awards 22-30 marks if the group project can be deployed as a commercial IoT application or service.

Practical use-case criterion	Project support
Real problem	energy waste, classroom safety and facility monitoring are real operational problems.
Deployable architecture	separate edge node, gateway and service layers can be maintained or replaced independently.
Commercial value	dashboard, logs, status indicators, mobile control and analytics resemble a facility operations product.
Cost feasibility	Arduino/ESP32 modules, low-cost sensors and a 12 V fan can be deployed as an affordable prototype.
User value	teachers or operators can use presets, mobile dashboard and voice commands instead of technical tools.
Safety value	local alarm handling avoids dependence on cloud services.
Expandability	MQTT topics support additional rooms, sensors or dashboards.

Why this targets the top band: It is not merely possible to imagine a real application. The prototype already behaves like a small commercial classroom IoT service: it monitors, decides, alarms, logs, accepts remote configuration and responds to natural-language commands.

4. Learning Outcomes

Outcome	Response
C: software and hardware in IoT	Arduino, ESP32, sensors, actuators, MQTT, dashboard and AI backend are all explained and implemented.
D: security and privacy of IoT	report covers local-first safety, key management, temporary audio, command validation and future hardening.
E: layered IoT architecture	the project explicitly separates physical, edge, gateway, service, intelligence and user layers.

Outcome	Response
F: IoT applications	the project implements a smart classroom energy-saving, safety and monitoring service with real hardware.

5. Suggested Examiner-Facing Summary

This project should be evaluated as a complete IoT application/service rather than as a single Arduino circuit. The system has a working edge controller, a network gateway, a local broker, a web operations dashboard, live MQTT telemetry, mobile voice control, AI task planning and physical actuator response. The GitHub repository provides code, documentation, wiring guides, self-check scripts and demonstration materials, showing both technical depth and operational maturity.